
BPM2_NB_ESOFT QUIZ 7_S2026

of Questions: 30

Question #: 1

A 65-year-old woman comes to the physician because of a 2-month history of ringing in her left ear. Her vital signs are within normal limits. Physical examination shows no abnormalities. An audiometry test shows deafness on the left side. A CT scan of the head shows an acoustic neuroma compressing the cochlear branch of the vestibulocochlear nerve at the internal acoustic meatus. Which of the following regions is most likely innervated by the affected nerve branch?

- A. Ciliary body
- ✓B. Organ of Corti
- C. Crista ampullaris
- D. Tympanic membrane
- E. Pinna of the ear
- F. Maculae

Rationale: The organ of Corti is innervated by the cochlear branch of CN8. The vestibular division/branch of the nerve receives information from the sensory organs of the vestibular labyrinth, which includes the crista ampullaris of the semicircular ducts and the maculae of the utricle and saccule.

Item Description: HCB.1382 - Innervation of the organ of Corti

Question #: 2

A 30-year-old woman comes to the physician for a scheduled thyroidectomy. She has a history of a malignant tumor of the thyroid gland diagnosed 6 weeks ago. Which of the following hormones is most likely secreted by the cells that normally line parenchymal follicles of the diseased organ?

- A. Aldosterone
- ✓B. Thyroxine
- C. Insulin
- D. Calcitonin
- E. Cortisol

Rationale: Under the influence of TSH, follicular cells secrete T3 and T4 hormones, which plays a role in body metabolism. They are called follicular cells because they are the cells that line the follicles of the thyroid gland.

Aldosterone and cortisol are hormones secreted by the adrenal gland. Insulin is secreted by the beta cells of the pancreas. Calcitonin is secreted by the parafollicular cells (C cells), found in the stroma of the thyroid gland.

Item Description: HCB.1040 - Hormones secreted by Thyroid Follicular cells

Question #: 3

A 1-month-old boy is brought to the physician because of concerns about his cleft lip and difficulties with breast feeding. Physical examination shows a cleft in the left side of the upper lip which continues a short distance into the anterior part of the hard palate. This condition most likely results from lack of fusion between which of the following structures?

- A. Medial and lateral nasal prominences
- ✓B. Intermaxillary segment and maxillary prominence of the first arch
- C. Intermaxillary segment and mandibular prominence of the first arch
- D. Lateral palatine process and maxillary prominence of the first arch
- E. Lateral palatine process and nasal septum

Rationale: The palate is formed by fusion between the intermaxillary segment and the secondary palate which is formed from the maxillary prominences of the first arch.

Item Description: A 1-month-old boy is

Question #: 4

A 70-year-old man comes to the emergency department because of sudden severe abdominal pain that began 1 hour back. His temperature is 37°C (98.6°F), pulse is 146/min, respirations are 17/min, and blood pressure is 170/90 mm Hg. His EKG shows atrial fibrillation. Abdominal examination shows diffuse tenderness on palpation. A CT arteriogram of the abdomen shows poor perfusion to a large segment of the jejunum. Blockage of which of the following vessels is most likely responsible for the patient's condition?

- A. Celiac trunk
- B. Inferior mesenteric artery
- ✓C. Superior mesenteric artery
- D. Superior rectal artery

E. Left gastric artery

Rationale: Small intestine that include lower part of duodenum, jejunum, ileum and transverse colon until the right two-third is from superior mesenteric artery

Item Description: ANAT.DG.1920 Superior mesenteric artery

Question #: 5

A 65-year-old-woman comes to the physician because of a 2 month history of intermittent bloody stools. Physical examination and imaging studies show a tumor of the distal GI tract. During surgical removal of the tumor, the inferior mesenteric, sacral, internal iliac and superficial inguinal lymph node groups are biopsied. Only the superficial inguinal lymph nodes are positive for cancerous cells. Which of the following parts of the GI tract is the most likely location of the tumor?

- A. Jejunum
- B. Ascending colon
- C. Rectum
- ✓D. Anal canal
- E. Descending colon

Rationale: medial group of the superficial inguinal lymphnodes receive the lymph from anal canal and perineum.

Item Description: ANAT.DG.2032 Innervation and lymphatic drainage of the anal canal

Question #: 6

A 25-year-old man comes to the emergency department because of a 2-day history of severe pelvic pain associated with fever and chills. His temperature is 38°C (100.4°F), pulse is 95/mins, blood pressure is 90/70mm Hg and respirations are 22/min. Imaging studies show an abscess in the deep perineal pouch. Which of the following best describes the location of this space?

- A. Between Colles' fascia and the perineal membrane
- B. Superior to the pelvic diaphragm and fascia
- ✓C. Between the perineal membrane and the pelvic fascia
- D. Surrounded by Bucks fascia
- E. Anterior to the testes in the scrotum

Rationale: deep perineal pouch is present between the perineal membrane and pelvic fascia

Item Description: ANAT.RP.0801 - deep perineal pouch

Question #: 7

An 8-year-old boy is brought to the physician by his mother for a follow-up examination. He has a life-long history of speech dysfluency and has just started speech therapy. He remains at home on all speech therapy days to avoid the embarrassment of leaving the school classroom for this therapy. Which of the following learning principles best accounts for his continued behavior of remaining at home on speech therapy days?

- A. Positive reinforcement
- ✓B. Negative reinforcement
- C. Positive punishment
- D. Negative punishment
- E. Classical extinction
- F. Operant extinction

Rationale: The following steps are a guide to answering operant questions:

1. What is the identified behavior?

The child remaining at home on speech therapy days

2. What is the consequence of this behavior?

Avoidance of embarrassment

3. Did the identified behavior increase or decrease following the consequence?

It increased (thus, the behavior has been reinforced)

4. Is the consequence of the identified behavior (avoidance of embarrassment) better described as "getting/applying" or "removing" something?

Removing (thus negative)

This is an example of negative reinforcement. The child's behavior of avoiding school (remaining at home) is strengthened because the consequence involved removing something unpleasant (embarrassment).

Recall that punishment and extinction processes result in the weakening of a behavior (not the strengthening of a behavior).

Item Description: ESoft.quiz.BEHS.0050.learning theory_operant_negative RF

Question #: 8

A 37-year-old man comes to the physician for a follow-up examination. His father recently died from Huntington's disease, and he has been discussing the possibility of being tested for the disease. He fully understands that he is at significant risk of developing the disease but decides that at this time in his life, he would rather not know whether he carries the gene. Which of the following defense mechanisms best explains the patient's decision not to be tested?

- A. Repression
- B. Denial
- C. Projection
- D. Intellectualization
- E. Rationalization
- ✓F. Suppression

Rationale: Suppression involves the voluntary, temporary removal from consciousness of some unpleasant or threatening future event. In this scenario the individual is consciously aware of the threatening aspect of reality (knowledge of susceptibility to the disease) but chooses not to think about it now. In contrast, repression involves the UNCONSCIOUS removal from consciousness of some threatening memory or event, or unacceptable drive.

Denial involves refusing to accept reality. In this case, the man is merely choosing not to know reality.

Item Description: Esoft.quiz.BEHS.0064.psychotherapies_defense mechanisms_suppression

Question #: 9

A 32-year-old man comes to the physician for a follow-up examination. He has a 2-year history of relationship problems and is starting psychotherapy. Throughout the psychotherapy session he gossips about his family members, friends, and neighbors. When describing his relationship with his sister, he says their relationship is fine even though she is "such a big gossip." Which of the following defense mechanisms best explains the patient's description of his sister?

- A. Reaction formation
- B. Displacement
- ✓C. Projection
- D. Identification
- E. Splitting
- F. Rationalization

Rationale: The man has the undesirable behavior of gossiping. He does not take ownership of this behavior but rather mirrors (projects) it on to somebody else (his sister). This is known as projection. Reaction formation is feeling inwardly one way and acting outwardly in the exact opposite way. Displacement is taking out one's feelings on somebody other than the source of those feelings. Identification (also known as introjection) is imitating the behaviors of somebody else because of dissatisfaction with oneself. Splitting is viewing the world in binary terms (all good, all bad). Rationalization is coming up with an explanation (a rationale) to justify your actions to counter one's feelings of guilt.

Item Description: Esoft.quiz.BEHS.0064.psychotherapies_defense mechanisms_projection

Question #: 10

A 72-year-old man comes to the physician for a follow-up examination. He has a 2-year history of gradual cognitive decline. Results from MRI and PET scans of the brain, APOE genetic testing, and neuropsychological evaluation are consistent with Alzheimer's dementia. The physician is reluctant to tell the patient the diagnostic bad news because the patient reminds the physician of his beloved grandfather. Which of the following psychological processes most likely contributes to the physician's reluctance to convey the bad news?

- A. Negative transference
- B. Positive transference
- C. Negative countertransference
- ✓D. Positive countertransference
- E. Denial

Rationale: Recall that the process of transference involves the patient reacting to the physician based on characteristics of her/his relationship with a significant person in their own life. In the case of countertransference, we see the physician doing the same thing in her/his interactions with the patient – that is, tending to relate to the patient as she/he had with a significant person in their own life. Accordingly, the case scenario is an example of positive countertransference because it involves the physician tending to be influenced in a positive (sentimental) way by similarities the patient bears with his grandfather. Even though the countertransference is characterized as "positive", it can have adverse effects on communication within the physician-patient relationship.

Item Description: Esoft.quiz.BEHS.0028.PPR_communication barriers_positive countertransference

Question #: 11

A 28-year-old man comes to the physician for a follow-up examination. He has a 10-year history of smoking cigarettes and has made the decision to stop smoking. He seeks the advice of his physician because he wants to start planning how to proceed with his intentions to stop smoking. Which of the following stages of behavior change is he most likely in?

- A. Precontemplation
- B. Contemplation
- ✓C. Preparation
- D. Action
- E. Maintenance

Rationale: The man has identified a health risk behavior (smoking) and made the decision to modify this behavior. Although he is still smoking, he has taken an initial step in changing his behavior by seeking the physician's assistance. This is the preparation stage.

Item Description: Esoft.quiz.BEHS.0037.PCI_stages of change_preparation

Question #: 12

A 47-year-old man comes to the physician for a routine examination. He is moderately overweight, and his cholesterol levels are elevated. The physician explains to him that his sedentary lifestyle and high-fat diet are contributing to his condition and recommends starting a diet and exercise program. The patient says he has tried similar approaches and has always failed because he simply does not have the determination to continue with it. Which of the following responses by the physician is most likely to motivate behavioral change in this patient?

- A. "Given the progressive nature of your condition you don't have much choice at this point."
- B. "It's your choice, of course, but you will need to accept responsibility for your decision."
- C. "There are several new dietary approaches available now. I'll bet you can find one you can stick to."
- ✓D. "I understand why it would be disheartening to have been unsuccessful in your earlier attempts, but would you consider working together on this?"
- E. "There are lots of others in your exact same situation. I'm wondering if you would consider joining the hospital weight management group?"

Rationale: Recall that hallmarks of motivational interviewing include empathy, curiosity, and openness on the part of the physician to assist in the behavior change process. Accordingly, the best response here is the key.

Item Description: ESoft.quiz.BEHS.BM.0038.PCI_motivational interviewing

Question #: 13

A 68-year-old woman is brought to the physician by her son because of a 2-month history of persistent headaches. She has not been using the prescribed medication given to her by the physician but instead has been taking herbal remedies from a tribal healer in her community. She said the herbal remedies have only provided minimal pain relief. Which of the following is the most appropriate initial response by the physician?

- A. "We will run a few more tests, and then I will give you a medicine that will work."
- B. "I don't need to know about the traditional remedies that you've tried. I need to know if you've taken any medicines from the drug store or pharmacy."
- C. "It is important that you refrain from these remedies while you are under my care."
- D. "I respect that you have your own cultural beliefs, but it's important that you don't let them interfere with my treatment recommendations."
- ✓E. "Please tell me about these remedies and how often you take them."

Rationale: It is important to learn as much as possible about alternative medical approaches that patients are using to determine whether there may be any interactions with allopathic treatment recommendations and to design a treatment plan that takes into account the patient's culture. Therefore, the best response for the physician is to ask an open-ended question, which would be "Please tell me

about these medications and how often you take them.”

Item Description: ESoft.quiz.BEHS.0027.PPR_respect culture_gather more info

Question #: 14

A 76-year-old woman comes to the physician because of a 2-week history of progressively worsening shortness of breath, swelling in both legs and increasing difficulty while climbing stairs. She has a 2-year history of congestive heart failure. She looks tired and anxious. Her respirations are 25/min. Physical examination shows coarse crackles at the base of her lungs and (2+) bilateral pedal edema up to her mid-calf. An X-ray of the chest shows bilateral pleural effusion. Which of the following components of her presentation are most likely objective and subjective findings, respectively?

A. Objective: coarse crackles at base of lung;

Subjective: (2+) bilateral pedal edema in lower limbs

B. Objective: coarse crackles at base of lung;

Subjective: respirations are 25/min

C. Objective: bilateral pedal edema in lower limbs;

Subjective: bilateral pleural effusion on x-ray of chest

✓D. Objective: (2+) bilateral pedal edema in lower limbs;

Subjective: progressively worsening shortness of breath

E. Objective: Progressively worsening shortness of breath;

Subjective: (2+) bilateral pedal edema in lower limbs

Rationale: Subjective findings ("symptoms") are those things the patient would tell you in their own words. For example: the progressively worsening shortness of breath, swelling in both legs, and shortness of breath that is worse with climbing stairs. A healthcare provider collects information about subjective findings during the history section of an encounter.

Objective findings ("signs") during physical examinations are found by the healthcare provider. For example: observing an anxious and tired-looking patient, the respiratory rate, coarse crackles at the bases of the lungs, (2+) bilateral edema. A healthcare provider collects information about objective findings during the physical examination section of an encounter.

The X-ray of the chest is neither a symptom nor a sign but rather a diagnostic tool used as part of the patient's evaluation to confirm a diagnosis. This is part of the investigations or diagnostic workup done during a patient encounter.

Question #: 15

A 16-year-old patient comes to the physician for a routine examination. The physician reviews the patient's chart ahead of time and notices that a typical man's name is listed and that the sex is indicated as male. However, the nurses refer to the patient as "she" and "her." Which of the following is the best initial question for the physician to ask during the medical encounter?

- A. "Should I address you as Mr. or Ms.?"
- B. "What is your sexual orientation?"
- C. "What is your gender identity?"
- ✓D. "What is your preferred name and pronoun?"
- E. "Is it okay for me to refer to you by your birth name?"

Rationale: The best response would be to ask the patient how they prefer to be addressed. Asking if "Mr. or Ms." is better or asking whether it is okay to call them by their birth name are close-ended questions. They include physician assumptions about appropriate options and do not encourage the patient to openly express their true preference. Asking open-ended questions about gender identity and sexual orientation will be important during the clinical interview but would not be the best questions to ask before initially addressing the patient.

Question #: 16

A 20-year-old woman is brought to the emergency department because of a 3-hour history of severe abdominal pain. She is disoriented and highly agitated. Physical examination shows diffuse abdominal tenderness. Urinalysis shows a purple color on exposure to air and light. She is treated for pain and her condition improves following a glucose infusion. Inherited deficiency of which of the following enzymes is most likely in this patient?

- A. Homogentisate oxidase
- B. Uroporphyrinogen decarboxylase
- C. Bilirubin UDP-glucuronyl transferase
- ✓D. Hydroxymethyl bilane synthase
- E. Cystathionine synthase
- F. Uroporphyrinogen-III synthase

Rationale: The vignette describes a patient with acute intermittent porphyria (AIP). Note the abdominal pain and presence of porphobilinogen in urine (Purple color of urine). The deficient enzyme is Hydroxymethyl bilane synthase (also known as porphobilinogen deaminase).

Item Description: BCHM.MB.2406; Acute intermittent porphyria (AIP)

Question #: 17

A 2-month-old girl is brought to the physician by her parents because of a 1-day history of intermittent seizures. The mother says the urine has a mousy (musty) odor. Physical examination shows light colored skin and a milestone developmental delay. She is prescribed a special diet lacking an amino acid. She is prescribed a supplement to facilitate metabolism of this amino acid. Which of the following supplements is most likely advised for this patient?

- A. Tetrahydrofolate
- ✓B. Tetrahydrobiopterin
- C. Pyridoxine (vitamin B6)
- D. Biotin
- E. Thiamine (vitamin B1)
- F. Cobalamin (Vitamin B12)

Rationale: The vignette describes a child with phenylketonuria. The musty odor of the urine is due to the presence of phenylacetate and phenyllactate in the urine. The enzyme deficient in this patient is phenylalanine hydroxylase. Phenylalanine hydroxylase requires tetrahydrobiopterin as a coenzyme.

Folate is required for purine and pyrimidine synthesis.

Pyridoxine supplements are useful in homocystinuria (activity of cystathionine synthase)

Biotin is required for carboxylases (Pyruvate carboxylase, propionyl CoA carboxylase and acetyl CoA carboxylase)

Thiamine supplements are useful in PDH (pyruvate dehydrogenase) deficiency and in maple syrup urine disease (Branched chain alpha-keto acid dehydrogenase deficiency).

Cobalamin (vitamin B12) supplements are useful in methylmalonic aciduria.

Item Description: PKU-quiz

Question #: 18

A 65-year-old woman comes to the physician because of a 2-month history of intermittent abdominal pain and weight loss. Physical examination shows jaundice and tenderness in the right upper quadrant of the abdomen. Laboratory studies are shown. Which of the following is the most likely diagnosis?

- A. Hemolytic anemia
- B. Acute viral hepatitis
- C. Chronic alcoholic cirrhosis
- ✓D. Obstructive jaundice
- E. Crigler Najjar syndrome type-II
- F. Gilbert syndrome

Rationale: The lab findings are characteristic of obstructive jaundice. Note the presence of conjugated (direct) hyperbilirubinemia, presence of bilirubin in urine. The absence of fecal stercobilin and urinary urobilinogen indicate complete obstruction of the biliary duct. Additional lab findings are elevated ALP and GGT levels.

Hemolytic anemia is characterized by elevated unconjugated (indirect) bilirubin levels.

In acute viral hepatitis, both direct and indirect bilirubin levels are generally elevated. Additional findings are bilirubin is found in urine. Serum ALT and AST levels are also elevated (ALT >>AST).

In chronic alcoholic cirrhosis, both direct and indirect bilirubin levels are elevated. Serum AST: ALT ratio is 2: 1. Serum albumin levels are low.

Crigler-Najjar syndrome and Gilbert syndrome are characterized by elevated unconjugated bilirubin levels. ALT and AST and ALP levels are within normal limits.

Attachment:

	Patient results	Reference results
Total serum bilirubin	6	1-2 mg/dL
Direct serum bilirubin	5.5	<0.2 mg/dL
Urine bilirubin	Present	Absent
Urine urobilinogen	Absent	Present
Fecal stercobilin	Absent	Present

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Item Description: Cholestatic jaundice bilirubin

Question #: 19

A 62-year-old man is brought to the emergency department in a comatose state. Physical examination shows jaundice and hepatomegaly. Laboratory studies show elevated serum bilirubin levels and a serum AST: ALT ratio is 2. He has a loss of recent memory, nystagmus, and difficulty walking, caused by a vitamin deficiency. Which of the following enzymes most likely requires this vitamin as a coenzyme?

- ✓A. Pyruvate dehydrogenase complex
- B. Methylmalonyl CoA mutase
- C. Lysyl oxidase
- D. Cystathionine synthase
- E. Methionine synthase

Rationale: The patient has chronic alcoholic cirrhosis. Note the jaundice and AST:ALT ratio of 2:1.

Thiamine deficiency in alcoholics presents as Wernicke Korsakoff syndrome.

Thiamine (Vitamin B1) is required for the activity of Pyruvate dehydrogenase, Alpha ketoglutarate dehydrogenase and branched chain ketoacid dehydrogenase complexes. It is also required for the activity of Transketolase (pentose phosphate pathway).

Item Description: BCHM.MB. 2803; Vit B1; Wernicke Korsakoff

Question #: 20

A 3-year-old girl is brought to the physician by her mother for a routine examination. The mother describes the child's recently developed habit of intentionally dropping her biscuits on the floor while sitting in her high chair at dinner. When this behavior occurs, the mother provides the child with more biscuits. The child seems to enjoy the attention involved in getting more biscuits, so her behavior of intentionally dropping biscuits on the floor has increased in frequency. The physician advises the mother to stop providing the child with more biscuits following this behavior. Which of the following learning principles is most likely the basis of the physician's advice?

- A. Positive reinforcement
- B. Negative reinforcement
- C. Positive punishment
- D. Negative punishment
- ✓E. Operant extinction

Rationale: Operant extinction involves no longer providing reinforcement following a previously reinforced behavior. In this scenario, the mother unintentionally reinforced the girl by rewarding her with attention (getting more biscuits) each time the girl dropped biscuits on the floor. The physician advises the mother to merely stop providing the reinforcer (attention/replacement biscuit) used to develop the biscuit-dropping behavior in the first place. This is the definition of operant extinction.

For the answer to be punishment, some other consequence must occur following the behavior. For example, if the mother yells at the girl for intentionally dropping her biscuits, then the answer would be positive punishment. If the mother took away the child's favorite toy when the child intentionally drops her biscuits, then this would be negative punishment.

The question is asking about the learning principle involved in trying to diminish the frequency of a behavior; thus, the answer cannot be reinforcement, which involves increasing a behavior following a consequence.

Item Description: ESoft.quiz.BEHS.0045.learning theory_operant_extinction

Question #: 21

A 51-year-old man comes to the physician because of a 2-month history of hearing loss in both ears. He has served in the military for the previous 30 years. An audiometry shows sensorineural hearing loss most likely caused by noise-induced trauma to the inner hair cells of the inner ear. Which of the following best characterizes the localization of these cells in the cochlea?

- A. Scala vestibuli
- ✓B. Scala media
- C. Scala tympani
- D. Semicircular canal
- E. Sacculle

Rationale: The hair cells involved in hearing are arranged in inner and outer rows of cells in the organ of Corti within the scala media or cochlear duct.

Item Description: HCB.1383 - Structure of the cochlea - scala media

Question #: 22

A 2-week-old boy is brought to the physician by his parents because of a 3-day history of constipation and vomiting. Physical examination shows an irritable baby, and a distended abdomen. Imaging studies of the abdomen show gut distension proximal to an area of constriction, and flattening of the gut distal to the same area of constriction. The physician suspects Hirschsprung's disease. This condition often occurs when there is absence of the myenteric plexus between the muscle layers of the gut wall. Which of the following cell bodies are most likely normally present in this plexus?

- A. Sensory nerve fibers
- B. Preganglionic parasympathetics
- ✓C. Postganglionic parasympathetics
- D. Preganglionic sympathetics
- E. Postganglionic sympathetics

Rationale: The myenteric (Auerbach's) plexus consists of the groups of parasympathetic ganglion cells and nerve fibers between the layers of the muscularis externa. These cells develop from neural crest cells that migrate to the tissue.

Sensory nerve fibers are usually found in the dorsal root and dorsal horn of the spinal cord with their cell bodies in the dorsal root ganglia. Preganglionic parasympathetic fibers are found in specific cranial nerves, or the lumbar and sacral regions of the spinal cord. Preganglionic sympathetic fibers are within the thoracic region of the spinal cord while the postganglionic sympathetic fibers are usually within the preaortic ganglia (egs celiac ganglia, superior cervical) or medulla of the adrenal gland.

Item Description: HCB.1265 - Hirschsprung disease - Myenteric plexus

Question #: 23

A 25-year-old man volunteers for a study related to the coronary blood flow under different physiologic conditions. He is instructed to perform graded exercise and his heart rate, blood pressure, coronary blood flow are measured. His resting heart rate is 84/min and his heart rate during moderate exercise is 120/min. Compared to at rest, which of the following options is most likely during a moderate exercise by this man?

Option	Diastolic duration	Coronary vascular resistance
A	Increased	Increased
B	Increased	Decreased
C	Decreased	Increased
D	Decreased	Decreased
E	Decreased	No change

- A. A
- B. B
- C. C
- ✓D. D
- E. E

Rationale: Coronary circulation adapts to the severity of exercise. During exercise, the heart rate increases due to sympathetic stimulation. Increase in heart rate decreases the cardiac cycle time and also duration of diastole. This may decrease the coronary blood flow as the blood flow is maximal during diastole. However, metabolic activity in the heart releases metabolites that cause coronary vasodilation. Coronary vasodilation drops (decreases) coronary vascular resistance which increases blood flow.

Item Description: PHYS.0148. Special Circulation

Question #: 24

A 30-year-old woman comes to the physician because of a 10-month history of substernal burning pain. She says the pain is worse in an empty stomach and is relieved after eating. Physical examination shows no abnormalities. Upper GI endoscopy shows erythematous mucosa extending from the gastroesophageal junction upward into the lower esophagus. She is diagnosed with gastroesophageal reflux disease with increased acid secretion in the stomach. Which of the following additional substances is most likely secreted by the cells that caused hypersecretion of acid in this patient?

- A. Mucus
- ✓B. Intrinsic factor
- C. Gastrin
- D. Somatostatin
- E. Bicarbonate

Rationale: In this patient, the acid production is increased. The parietal (oxyntic) cells are the source of both acid and intrinsic factor.

Item Description: PHYS.1123. Secretory products of the parietal cells of the stomach

Question #: 25

A 39-year-old woman comes to the physician because of a 9-week history of being 'emotionally labile' and 'restless.' Her temperature is 37.8°C (100°F), pulse is 124/min, blood pressure is 138/72 mm Hg, and respirations are 20/min. Physical examination shows exophthalmos, pretibial myxedema, and fine tremors of outstretched hands. Laboratory studies show elevated serum thyroid hormones and an autoantibody against TSH receptors. Which of the following additional laboratory alterations is most likely in this patient?

- A. Elevated serum TSH
- B. 2. Elevated serum TRH
- ✓C. Low serum TSH
- D. Elevated serum cholesterol
- E. Elevated serum catecholamines

Rationale: This is a case of Grave's disease, an autoimmune disease. The autoantibodies against TSH receptors stimulate thyroid gland to secrete increased T3, T4. This suppresses anterior pituitary and hypothalamus. This results in reduced secretion of TSH and TRH from anterior pituitary and hypothalamus respectively.

Item Description: PHYS.1055. Graves disease Lab findings

Question #: 26

A 24-year-old man comes to the physician because of increased thirst and frequent micturition that have lasted for 2 weeks. He was discharged from hospital 3 weeks ago after being managed for head injury he sustained in a car crash. His vital signs are all within normal limits. Physical examination shows no abnormalities. Laboratory studies show a dilute urine. Which additional laboratory finding is most likely present in this patient?

- A. Decreased cortisol
- B. Increased cortisol
- C. Increased thyroxine
- ✓D. Decreased vasopressin
- E. Increased prolactin

Item Description: SOM.MK.1.BPM2.1.NB.1.NSCI.0278 Hypothalamus and Pituitary (ANS)

Question #: 27

A 62-year-old man comes to the physician because of a 6-hour history of nausea and vomiting. MRI of the head shows a lesion affecting the left vestibular labyrinth. Which of the following outcomes is most likely?

- A. Persistent deviation of the eyes to the left
- B. Persistent deviation of the eyes to the right
- ✓C. Slow leftward drift of the eyes followed by fast rightward resetting
- D. Slow rightward drift of the eyes followed by fast leftward resetting
- E. Leftward drift of the left eye
- F. Rightward drift right eye

Rationale: With a lesion to the left vestibular labyrinth, the right vestibular labyrinth is unopposed, driving both eyes slowly to the left. The left frontal lobe then rapidly resets the eyes to the right.

Item Description: SOM.MK.II.BPM2.1.NB.4.NSCI.0253 Pathological nystagmus

Question #: 28

A 26-year-old woman at 28 weeks' gestation comes to the physician for a routine antenatal examination. Her vital signs are within normal limits. Physical examination shows height of the uterine fundus corresponding to 3 fingerbreadths above the umbilicus and normal fetal heart sounds. A decrease in which of the following hormones is most likely in her?

- A. Cortisol
- B. Adrenocorticotrophic hormone (ACTH)
- ✓C. Luteinizing hormone
- D. Prolactin
- E. Parathyroid hormone (PTH)

Rationale: All hormones listed in the options are elevated during this stage of pregnancy EXCEPT luteinizing hormone. LH being a gonadotropin from anterior pituitary is inhibited by elevated estrogen and progesterone, secreted from the placenta.

Item Description: PHYS.1087.Pregnancy Hormone Levels LH

Question #: 29

A 61-year-old woman is brought to the emergency department because of a 20-minute history of unconsciousness. MRI of the head shows a left epidural hemorrhage resulting in bilateral compression of the midbrain. Which of the following outcomes is most likely upon the infusion of warm water into the left ear of the patient?

- A. No response
- ✓B. Right eye deviates right
- C. Both eyes deviate right
- D. Left eye deviates right
- E. Right eye deviates left
- F. Both eyes deviate left
- G. Left eye deviates left

Rationale: Stimulation of the left labyrinth should drive the right eye to the right via activation of the right abducens nerve. The motor nucleus of the abducens nerve also ought to activate the left oculomotor nucleus, resulting in medial deviation of the left eye. However, with compression of the midbrain, the left medial rectus muscle should remain uncontracted, leading to failed adduction of the left eye.

Item Description: SOM.CS.I.BPM2.1.NB.4.NSCI.0252 Caloric and oculocephalic testing in comatose patient

Question #: 30

A 70-year-old woman is brought to the physician by her son because of a six-week history of difficulty emptying her urinary bladder completely. She feels embarrassed because of her recent incidences of urinary dribbling in public places. Her vital signs are normal. Her brain imaging studies indicate a central lesion leading to her neurological problem. Which of the following structures is most likely damaged in this patient?

- A. Anterior hypothalamus
- ✓B. Pontine areas
- C. Medullary reticular formation
- D. Posterior hypothalamus
- E. Subfornical organ

Rationale: Hints: The pontine micturition center regulates the bladder micturition reflex. Damage to this portion of the pons could account for the patient's inability to empty the bladder completely, leading to urinary dribbling. This is also known as automatic bladder.

Item Description: Automatic bladder